

CMT 451 – Mobile Security

Course Description

This course explores security challenges in mobile computing environments, focusing on mobile operating systems, applications, networks, and international cybersecurity standards.

Course Objectives / Learning Outcomes (CBET)

- Identify mobile security threats and vulnerabilities
- Design and evaluate secure mobile systems
- Apply international cybersecurity standards to mobile environments

Course Structure and Standards Mapping (6 Modules)

Module 1: Mobile Computing and Threat Landscape

Indicative Content:

- Mobile architectures
- Threat actors
- Attack surfaces

Mapped Standards / Frameworks:

- NIST CSF – Identify
- ISO/IEC 27001 Annex A

Module 2: Mobile Operating Systems Security

Indicative Content:

- Android security
- iOS security
- OS-level controls

Mapped Standards / Frameworks:

- NIST SP 800-53
- ISO/IEC 27001 Annex A – Access control

Module 3: Mobile Application Security

Indicative Content:

- Secure coding
- Data storage
- Secure communication

Mapped Standards / Frameworks:

- OWASP MASVS
- ISO/IEC 27001 Annex A – Cryptography

Module 4: Mobile Malware and Attack Techniques

Indicative Content:

- Mobile malware
- Reverse engineering
- Attack mitigation

Mapped Standards / Frameworks:

- OWASP MASVS – Resilience
- NIST CSF – Detect

Module 5: MDM, Forensics, and Incident Response

Indicative Content:

- Mobile device management
- Forensics basics
- Incident response

Mapped Standards / Frameworks:

- NIST SP 800-61
- ISO/IEC 27001 Annex A – Incident management

Module 6: Emerging Trends and Secure Mobile Design

Indicative Content:

- IoT and mobile
- Privacy
- Secure-by-design principles

Mapped Standards / Frameworks:

- NIST SSDF
- ISO/IEC 27001 Annex A – Secure operations

List of Assessments and Marks

Student performance in this course shall be evaluated through a combination of continuous assessment and a final examination, structured as follows:

- **Continuous Assessment Tests (CATs) and Assignments: 30%**
These shall assess students' understanding of theoretical concepts, practical application of skills, problem-solving ability, and alignment with prescribed standards and frameworks covered in the course.
- **Final Examination: 70%**
The final examination shall evaluate comprehensive knowledge of the course content, including conceptual understanding, analytical ability, and application of principles in real-world and professional contexts.

Key References (APA)

Stallings, W. (2020). *Cryptography and network security* (8th ed.). Pearson.

OWASP Foundation. (2023). *Mobile Application Security Verification Standard (MASVS)*.

NIST. (2018). *Framework for Improving Critical Infrastructure Cybersecurity*.

ISO/IEC. (2022). *ISO/IEC 27001: Information security management systems*.